Day 15: Model selection

Warm up (quiz review)

`bikes_A` predicts `rides` from `temp_actual`, `windspeed`, and `weekend`. `bikes_B` adds `holiday`.

```
anova(bikes_A, bikes_B)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend
Model 2: rides ~ temp_actual + windspeed + weekend + holiday
  Res.Df       RSS Df Sum of Sq      F   Pr(>F)
1    496 755019445
2    495 743486506  1  11532939 7.6784 0.005799 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

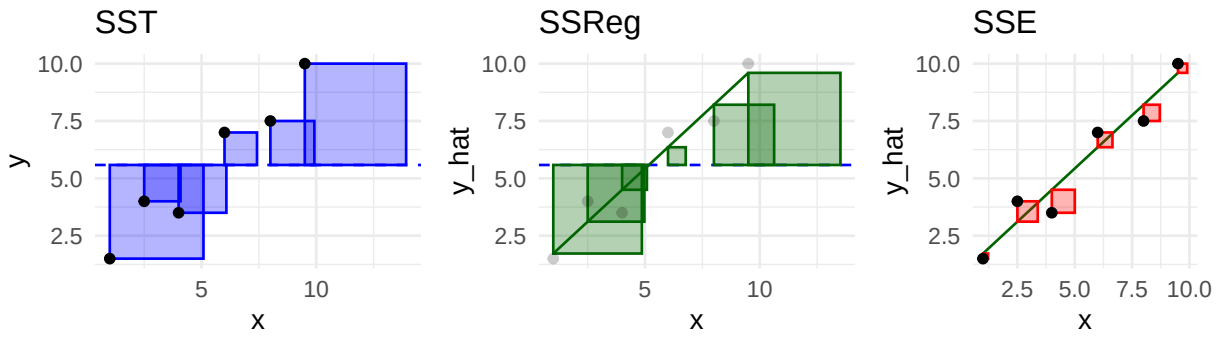
1. Which model is the full model? Write the null and alternative hypothesis for the test in the ANOVA table above.

2. What is d? What are df_{full} and $SSE_{reduced}$?

3. State your conclusion in context.

4. What p-value would you expect for the `holidayyes` row of `summary(bikes_B)`? Why?

Review: sums of squares



- Least squares makes the total area of the _____ squares as small as possible
- R^2: proportion of the _____ area (SST) covered by the _____ squares (SSReg)
- $R^2_{adj} = 1 - \frac{MSE}{MST}$: same idea, but each model is "charged" for the number of _____ it uses
- Overall F test, $F = \frac{MSReg}{MSE}$: is the "average" _____ square bigger than the "average" _____ square?
- Extra sums of squares F test: are the _____ squares for a larger model significantly smaller than for a smaller model?

Today: building an explanatory model

Question: How are weekends and holidays associated with ridership, after accounting for the weather?

Two reasons to build a model:

- **Explanatory:** _____
- **Predictive:** _____

Today we're building an explanatory model. (Predictive models come after the quiz!)

Which variables would you include to answer our question? Why?

A recipe for explanatory models

1. **Start with the question.** Which variables are you interested in? Which do you need to adjust for? These stay in the model, even if _____
2. **Look before you fit.** Check for predictors that _____
3. **Fit a *small* set of candidate models** (not every possible combination!)

4. **Compare the candidates**: _____ for nested models; _____ for any models
5. **Check the conditions** for the model you choose
6. **Interpret and justify** your choice

Step 1: Our candidate models

- **Interested in:** weekend, `holiday`
- **Adjust for:** weather (temperature, windspeed, humidity)

Model	Predictors
A	`temp_actual + windspeed + weekend`
B	A + `holiday`
C	B + `humidity`
D	C + `temp_feel`
E	`temp_feel + humidity + weekend + holiday`

Step 2: Look before you fit

```
bikes |>
  select(temp_actual, temp_feel, windspeed, humidity) |>
  cor() |>
  round(2)
```

```
            temp_actual temp_feel windspeed humidity
temp_actual        1.00      0.98     -0.11     0.27
temp_feel          0.98      1.00     -0.15     0.27
windspeed         -0.11     -0.15      1.00    -0.32
humidity           0.27      0.27     -0.32     1.00
```

Which pair of predictors could cause a problem? What problem?

Step 3: Compare nested models with an F test

Two models are **nested** if the reduced model is a _____ of the full model (we get it by setting some β's equal to _____).

Which pairs are nested? Circle one for each.

Model 1	Model 2	Nested?
A	B	yes / no
B	D	yes / no

Model 1	Model 2	Nested?
C	E	yes / no
A	C	yes / no

$$F = \frac{(SSE_{reduced} - SSE_{full})/d}{SSE_{full}/df_{full}}$$

- Numerator: _____
- Denominator: _____
- A large F means: _____

Adding variables one at a time:

step	d	F	p-value
A to B: add holiday	1	7.68	0.006
B to C: add humidity	1	85.43	<0.001
C to D: add temp_feel	1	3.28	0.071

What do the F tests say about each step?

Careful: `temp_actual` and `temp_feel`

model	term	estimate	std.error	p.value
C (without temp_feel)	temp_actual	94.88	5.21	<0.001
D (with temp_feel)	temp_actual	46.31	27.30	0.090
D (with temp_feel)	temp_feel	45.07	24.87	0.071

What happened to the standard error of `temp_actual`? Why? What should we do?

When the F test can't help

Is `bikes_E` **nested in** `bikes_C`**?** _____ Can we use `anova(bikes_E, bikes_C)`? _____

Two situations where the F test can't help:

1. _____
2. _____

We need tools that work for *any* two models fit to the same data.

Adjusted R^2

$$R^2_{adj} = 1 - \frac{MSE}{MST} = 1 - \frac{SSE/(n-(p+1))}{SST/(n-1)}$$

- R^2 always goes up when we add predictors; R^2_{adj} _____
- can be used for nested **and** non-nested models
- no formal test or rule for "better enough"

Model	Adj R2
C: temp_actual + windspeed + weekend + holiday + humidity	0.483
E: temp_feel + humidity + weekend + holiday	0.445

Which model does adjusted R^2 prefer? _____

Information criteria: AIC and BIC

$$IC = \text{Measure of Fit} + \text{Penalty}$$

$$AIC = n\log(SSE/n) + 2k \qquad BIC = n\log(SSE/n) + \log(n)\cdot k$$

where k = number of coefficients (**including** the intercept). In R: `AIC(model)` and `BIC(model)`.

- Smaller values are _____.
- When we add a predictor, the fit term gets _____ and the penalty gets _____.
- BIC penalizes more than AIC whenever $\log(n) >$ _____, i.e. $n \geq$ _____.

Example: adjusted R^2 and F tests by hand

A researcher fits four nested models to $n = 40$ observations. Fill in the table, then answer the questions.

Model	Predictors	p	SSE	$\hat{\sigma}^2 = SSE/(n-(p+1))$	R^2_{adj}
M0	none	0	980		
M1	x_1	1	520		
M2	x_1, x_2	2	410		
M3	x_1, x_2, x_3	3	398		

$R^2_{adj} = 1 - \hat{\sigma}^2/\hat{\sigma}^2_{M0}$ (since $MST = SSE_{M0}/(n-1)$).

(a) Which model is best by R^2_{adj}?

(b) Use an F test to compare M2 to M1 and M3 to M2. State d and df_{full} each time.

(c) Do the two tools agree? Which model would you choose?

Step 4: Compare all of our candidates

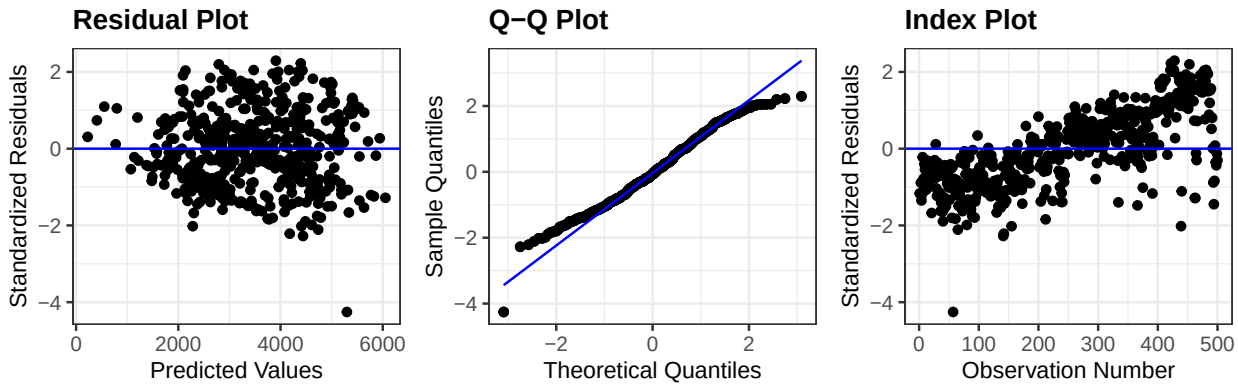
> **Activity**
>
> Work through Parts 1-3 of the R activity and record your results here.

Model	k	R^2_{adj}	AIC	BIC
A: `temp_actual + windspeed + weekend`				
B: A + `holiday`				
C: B + `humidity`				
D: C + `temp_feel`				
E: `temp_feel + humidity + weekend + holiday`				

Which model would you choose to answer our question? Justify your choice.

> **Stop here**
>
> Let Amanda know when your group finishes!

Steps 5 and 6: Check and interpret

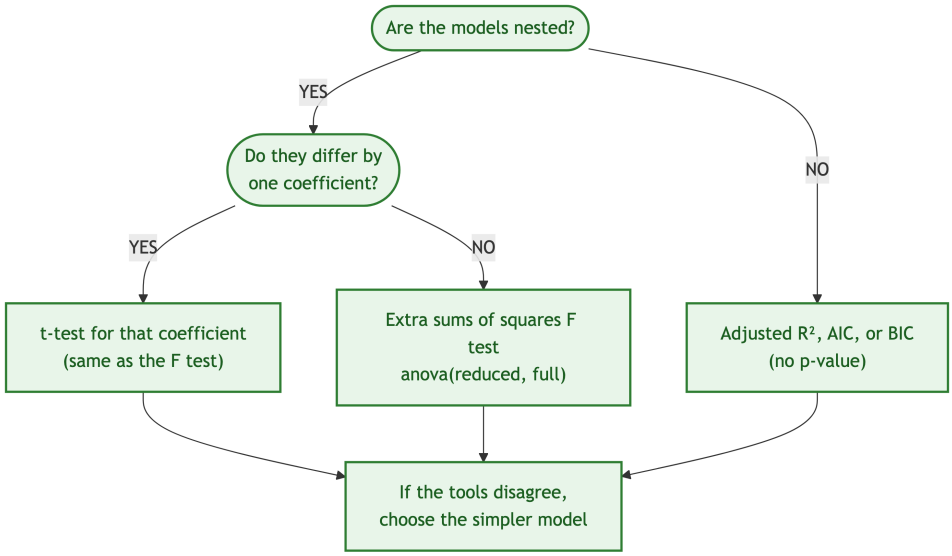


term	estimate	conf.low	conf.high
weekendTRUE	-777	-1000	-554
holidayyes	-1093	-1751	-435

term	estimate	conf.low	conf.high

Example: Using model C, answer our question in context.

Comparing two models



Summary

Tool	Nested only?	Gives a p-value?	Notes
F test			
Adjusted R^2			
AIC			
BIC			

- Use domain knowledge first. These tools compare candidate models; they don't replace thinking.
- **Justify your model choice!**